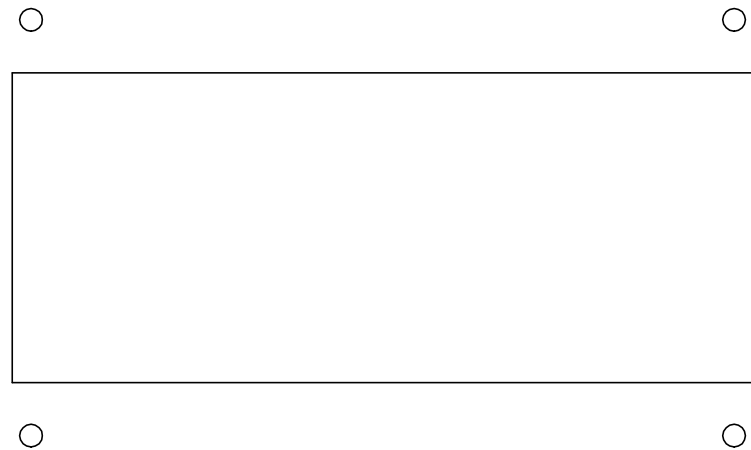




TRƯỜNG ĐẠI HỌC CÔNG NGHỆ KỸ THUẬT TP. HCM
Ho Chi Minh City University of Technology and Engineering
KHOA GIAO THÔNG VÀ NĂNG LƯỢNG
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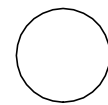
BALANCE



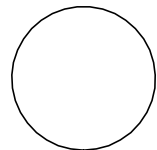
ON

OFF

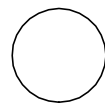
LEFT



POWER



RIGHT



LOAD



ON

OFF

UART



ON

OFF

LCD MODE



SETTING

MONITOR

CURRENT
DISCHARGE
MAX



VOLTAGE
DISCHARGE
MIN



CURRENT
CHARGE
MAX



VOLTAGE
CHARGE
MAX



DESIGN AND DEVELOPMENT OF A BATTERY MONITORING AND WARNING SYSTEM ON AN EMBEDDED SYSTEM PLATFORM USING CAN BUS

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Dr. Le Thanh Phuc

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SYSTEM SPECIFICATIONS

MCU: STM32F103C8T6 (Cortex-M3)
Network: 1 Master + 2 Slaves, CAN 2.0A @125kbps
Battery: Li-ion 18650, 10S2P (2x5S2P); 30 - 42V, nom. 36V
Monitoring: Cell V, Total V, Vmin/Vmax/Delta, Current, Temp, SOC
Balancing: Passive
Protection: OV / UV / OC / OT
Display: 20x4 LCD (I2C)
Alert: LED + Buzzer
Telemetry: CAN + UART